

6.11 Projection in the population: best linear prediction

So far the geometry has lived in observation space $\mathbb{R}^n$, and every statement was about one fixed data vector. There is a parallel geometry one level up, in which the vectors are *random variables* and the inner product is an expectation. In that geometry, least squares estimates something even when the linear model is false, namely the population projection of Y onto a span of regressors. The conditional expectation turns out to be a projection too. This section sets up the correspondence. Chapter 14 develops prediction theory from it, and Chapter 19 uses it to study what least squares estimates under misspecification.

6.11.1 Random variables as vectors

Consider random variables with finite second moments on a common probability space. They form a vector space, and

$$\langle U, V \rangle = \mathbf{E}(UV), \quad \|U\| = (\mathbf{E}U^2)^{1/2}$$

is an inner product, provided we identify variables that are equal with probability one, so that $\|U\| = 0$ forces $U = 0$. This space is usually written L^2 . Two random variables are orthogonal when $\mathbf{E}(UV) = 0$. For variables with mean zero, this is the same as being uncorrelated. The Cauchy–Schwarz inequality Eq. 6.2.2 becomes the statement that correlations lie in $[-1, 1]$. The *cosine* of the angle between two centred random variables is their correlation.

For a random k -vector $\mathbf{X}$ with finite second moments, the set $\mathcal{L} = \{a + \mathbf{bT}\mathbf{X} : a \in \mathbb{R}, \mathbf{b} \in \mathbb{R}^k\}$ is a subspace of dimension at most $k + 1$. The proofs of Proposition 6.2.2, Theorem 6.2.3 and Theorem 6.3.1 used only the axioms of an inner product and a finite spanning set, so they apply unchanged. Every Y with $\mathbf{E}Y^2 < \infty$ has a unique nearest point in $\mathcal{L}$, characterized by orthogonality of the error to $\mathcal{L}$.

Definition 6.11.1 — Best linear predictor

The **best linear predictor** of Y given $\mathbf{X}$, written $L(Y | \mathbf{X})$, is the projection of Y onto $\mathcal{L}$. It is the $a + \mathbf{bT}\mathbf{X}$ that minimizes the mean squared error $\mathbf{E}(Y - a - \mathbf{bT}\mathbf{X})^2$.

Theorem 6.11.1

Let $\boldsymbol{\mu}_X = \mathbf{E}\mathbf{X}$, $\mu_Y = \mathbf{E}Y$, $\boldsymbol{\Sigma}_{XX} = \text{Cov}(\mathbf{X})$ and $\boldsymbol{\sigma}_{XY} = \text{Cov}(\mathbf{X}, Y)$, and suppose $\boldsymbol{\Sigma}_{XX}$ is positive definite. Then

$$L(Y | \mathbf{X}) = \alpha^* + \boldsymbol{\beta}^{*\top} \mathbf{X}, \quad \boldsymbol{\beta}^* = \boldsymbol{\Sigma}_{XX}^{-1} \boldsymbol{\sigma}_{XY}, \quad \alpha^* = \mu_Y - \boldsymbol{\beta}^{*\top} \boldsymbol{\mu}_X.$$

The prediction error $U = Y - L(Y | \mathbf{X})$ has mean zero and is uncorrelated with every component of $\mathbf{X}$. Moreover

$$\text{Var}(Y) = \text{Var}(L(Y | \mathbf{X})) + \mathbf{E}U^2, \quad \frac{\text{Var}(L(Y | \mathbf{X}))}{\text{Var}(Y)} = \frac{\boldsymbol{\sigma}_{XY} \mathbf{T} \boldsymbol{\Sigma}_{XX}^{-1} \boldsymbol{\sigma}_{XY}}{\text{Var}(Y)}. \quad (6.11.1)$$

Proof. Orthogonality of $U = Y - a - \mathbf{b} \mathbf{T} \mathbf{X}$ to the spanning set $\{1, X_1, \dots, X_k\}$ of $\mathcal{L}$ gives the *population normal equations*

$$\mathbf{E}(U) = 0, \quad \mathbf{E}(\mathbf{X}U) = \mathbf{0}.$$

The first gives $a = \mu_Y - \mathbf{b} \mathbf{T} \boldsymbol{\mu}_X$. Substituting into the second and subtracting $\boldsymbol{\mu}_X \mathbf{E}(U) = \mathbf{0}$ gives $\text{Cov}(\mathbf{X}, Y) - \text{Cov}(\mathbf{X}) \mathbf{b} = \mathbf{0}$, so $\mathbf{b} = \boldsymbol{\Sigma}_{XX}^{-1} \boldsymbol{\sigma}_{XY}$. By the projection theorem the solution is the unique minimizer. Since U has mean zero and is uncorrelated with $\mathbf{X}$, it is uncorrelated with $L(Y | \mathbf{X})$, so the variances add. Finally, $\text{Var}(\boldsymbol{\beta}^{*\top} \mathbf{X}) = \boldsymbol{\beta}^{*\top} \boldsymbol{\Sigma}_{XX} \boldsymbol{\beta}^* = \boldsymbol{\sigma}_{XY} \mathbf{T} \boldsymbol{\Sigma}_{XX}^{-1} \boldsymbol{\sigma}_{XY}$. ■

The ratio in Eq. 6.11.1 is the population *squared multiple correlation*, the population counterpart of R^2 . By the same cosine argument as Theorem 6.7.2, it is the squared correlation between Y and $L(Y | \mathbf{X})$.

6.11.2 Conditional expectation is also a projection

The linear span $\mathcal{L}$ is one subspace of L^2 . A much larger one is $\mathcal{G} = \{g(\mathbf{X}) : \mathbf{E}g(\mathbf{X})^2 < \infty\}$, all square-integrable functions of $\mathbf{X}$. It is infinite-dimensional, but it is closed, and the projection theorem still holds in it. We accept that fact from measure-theoretic probability and only use its conclusion.

Proposition 6.11.2

Let $m(\mathbf{X}) = \mathbf{E}(Y | \mathbf{X})$ with $\mathbf{E}Y^2 < \infty$. Then:

- (a) $m(\mathbf{X})$ is the projection of Y onto $\mathcal{G}$. It minimizes $\mathbf{E}(Y - g(\mathbf{X}))^2$ over g , and $\mathbf{E}[(Y - m(\mathbf{X}))g(\mathbf{X})] = 0$ for all $g(\mathbf{X}) \in \mathcal{G}$;
- (b) $L(Y | \mathbf{X}) = L(m(\mathbf{X}) | \mathbf{X})$: the best linear predictor of Y equals the best linear predictor of the regression function;
- (c) $\mathbf{E}(Y - L(Y | \mathbf{X}))^2 = \mathbf{E}(Y - m(\mathbf{X}))^2 + \mathbf{E}(m(\mathbf{X}) - L(Y | \mathbf{X}))^2$;
- (d) if m is affine, $m(\mathbf{X}) = L(Y | \mathbf{X})$.

Proof.

- (a) The orthogonality property follows from the tower rule: $\mathbf{E}[(Y - m(\mathbf{X}))g(\mathbf{X})] = \mathbf{E}[g(\mathbf{X}) \mathbf{E}(Y - m(\mathbf{X}) | \mathbf{X})] = 0$. Orthogonality to the subspace characterizes the nearest point, by the argument of Theorem 6.3.1, which needs only Pythagoras.
- (b) This is Theorem 6.5.1(a) in L^2 . Since $\mathcal{L} \subseteq \mathcal{G}$, projecting onto $\mathcal{L}$ can be done in two stages.
- (c) $Y - L = (Y - m) + (m - L)$, and $m - L \in \mathcal{G}$ is orthogonal to $Y - m$ by (a).

(d) If $m \in \mathcal{L}$, the second term in (c) is zero. ■

Part (c) is the population version of Figure 6.5.1. The error of the best linear predictor splits into two orthogonal parts. One is irreducible noise, which no function of $\mathbf{X}$ can remove. The other is *approximation error*, the price of insisting on linearity. Chapters in Part IX reduce the second part by enlarging $\mathcal{L}$ towards $\mathcal{G}$.

6.11.3 Least squares estimates the population projection

Here is the connection with the rest of the chapter. Given data $(\mathbf{x}_{(i)}, y_i)$, put probability $1/n$ on each row. For this *empirical distribution*, the L^2 inner product of two variables is $n^{-1} \sum_i u_i v_i = n^{-1} \mathbf{u}^T \mathbf{v}$, which is the Euclidean inner product of observation space up to the factor $1/n$. So the best linear predictor under the empirical distribution is the least squares fit, and the population normal equations become the normal equations Eq. 6.4.1. Observation-space geometry is L^2 geometry for the empirical distribution.

As the sample grows, the empirical distribution approaches the population distribution, and the projections converge with it.

Proposition 6.11.3 — Consistency for the projection

Let $(\mathbf{X}_i, Y_i)$, $i = 1, 2, \dots$, be independent copies of $(\mathbf{X}, Y)$ with finite second moments and Σ_{XX} positive definite. Let $(\hat{\alpha}_n, \hat{\beta}_n)$ be the least squares coefficients from the first n observations, with an intercept. Then $(\hat{\alpha}_n, \hat{\beta}_n) \rightarrow (\alpha^*, \beta^*)$ with probability one. No assumption about the form of $\mathbf{E}(Y | \mathbf{X})$, the distribution of the errors, or their variance is needed.

Proof. For large n , $\hat{\beta}_n = \mathbf{S}_{XX}^{-1} \mathbf{s}_{XY}$, where $\mathbf{S}_{XX}$ and $\mathbf{s}_{XY}$ are sample covariances with divisor n (Section 6.6, centring). By the strong law of large numbers the sample means, second moments and cross-moments converge almost surely to their population values. So $\mathbf{S}_{XX} \rightarrow \Sigma_{XX}$ and $\mathbf{s}_{XY} \rightarrow \sigma_{XY}$. Matrix inversion is continuous at the nonsingular Σ_{XX} , so eventually $\mathbf{S}_{XX}$ is invertible and $\hat{\beta}_n \rightarrow \beta^*$. The intercept follows the same way. ■

Example. (A curved mean)

Let $X \sim \text{Uniform}(0, 2)$ and $Y = e^X + \varepsilon$ with independent $\varepsilon \sim N(0, 0.25)$. The regression function is not linear, but Theorem 6.11.1 still identifies a best line. A short calculation gives $\text{Cov}(X, Y) = 1$ and $\text{Var}(X) = 1/3$, so the population projection has slope 3.000 and intercept 0.195. The listing fits least squares lines to samples of increasing size. The estimates settle on these values, not on any feature of e^x . Figure 6.11.1 shows the regression function, the population projection, and a least squares fit.

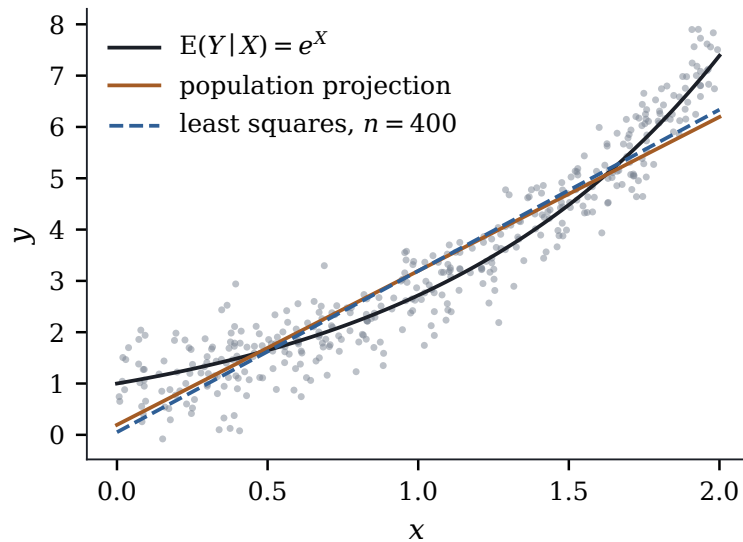


Figure 6.11.1: When the regression function is curved, least squares estimates its best linear approximation in the L^2 sense, the population projection. The approximation error is largest where e^x bends away from the line.

```
import numpy as np
rng = np.random.default_rng(611)

# X ~ Uniform(0, 2), E(Y | X) = exp(X), Var(Y | X) = 0.25.
# Population projection of Y onto span{1, X}: slope Cov(X, Y)/Var(X), intercept E Y
# - slope * E X.
EX, VarX = 1.0, 4.0 / 12.0
EY = (np.exp(2) - 1) / 2
EXY = (np.exp(2) + 1) / 2
slope_pop = (EXY - EX * EY) / VarX
intercept_pop = EY - slope_pop * EX

# E exp(X)
# E X exp(X)

for n in [20, 200, 2000, 20000]:
    x = rng.uniform(0, 2, n)
    y = np.exp(x) + rng.normal(0, 0.5, n)
    b, *_ = np.linalg.lstsq(np.column_stack([np.ones(n), x]), y, rcond=None)
    print(f"n = {n:6d}    intercept {b[0]:.3f}    slope {b[1]:.3f}")
print(f"population    intercept {intercept_pop:.3f}    slope {slope_pop:.3f}")
```

[Run this code online](#)

Remark (What changes under misspecification). [Proposition 6.11.3](#) is reassuring about the *target*. A least squares line always estimates a well-defined population quantity. It says nothing about the *uncertainty* of the estimate. When $\mathbf{E}(Y | \mathbf{X})$ is not linear, or the variance is not constant, the approximation error $m(\mathbf{X}) - L(Y | \mathbf{X})$ behaves like extra noise that depends on $\mathbf{X}$. The textbook formula $\sigma^2(\mathbf{X}^T\mathbf{X})^{-1}$ for the covariance of $\hat{\beta}$ is then wrong, even in large samples. The “sandwich” covariance estimators of Chapter 21 are designed for exactly this situation.